

Two variable K maps (F(A,B))

① $F = \sum m(0,1)$

A \ B	0	1
0	1	1
1		

$$\Rightarrow F(A,B) = \bar{A}$$

② $F = \sum m(1,2)$

A \ B	0	1
0		1
1		1

$$= \bar{A}B + AB$$

③ $F = \sum m(0,3)$

A \ B	0	1
0	1	
1		1

$$= \bar{A}\bar{B} + AB$$

④ $F = m(1) + d(0,2)$

A \ B	0	1
0	X	1
1	X	

$$= F(A,B) = \bar{A}$$

⑤ $F = \sum m(0,1,2)$

A \ B	0	1
0	1	1
1	1	

$$= \bar{A} + \bar{B}$$

⑥ $F = \sum m(3) + d(1)$

A \ B	0	1
0		X
1		1

$$= B$$

⑦ $F = \sum m(0,2)$

A \ B	0	1
0	1	
1	1	

$$= \bar{B}$$

⑧ $F = \sum m(1,3)$

A \ B	0	1
0		1
1		1

$$= B$$

⑨ $F = \sum m(2) + d(0,3)$

A \ B	0	1
0	X	
1	1	X

$$= A / \bar{B}$$

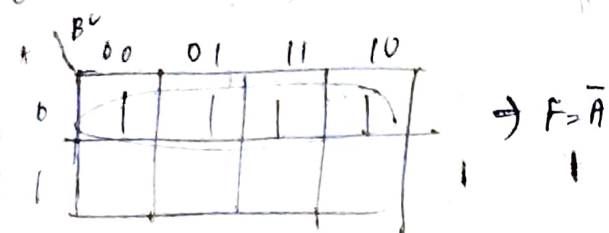
⑩ $F = \sum m(0,1,2,3)$

A \ B	0	1
0	1	1
1	1	1

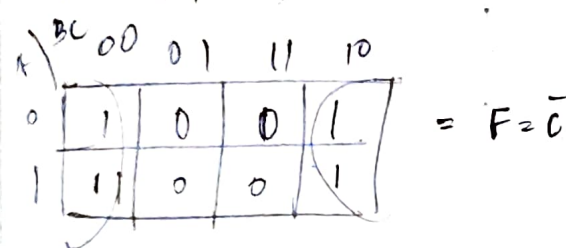
$$= F = 1 //$$

Three variable k maps (F(A,B,C))

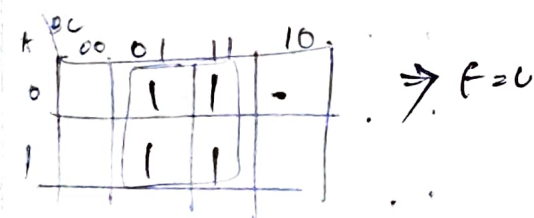
1) $F = \sum m(0,1,2,3)$



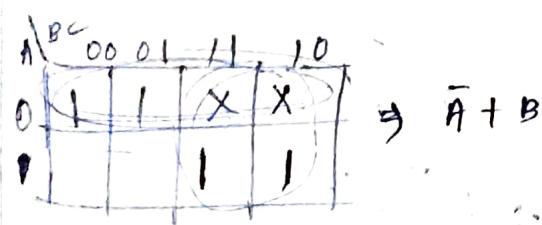
2) $F = \sum m(0,2,4,6)$



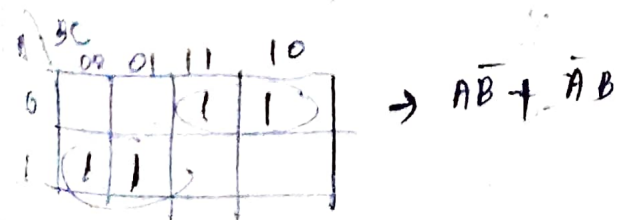
3) $F = \sum m(1,3,5,7)$



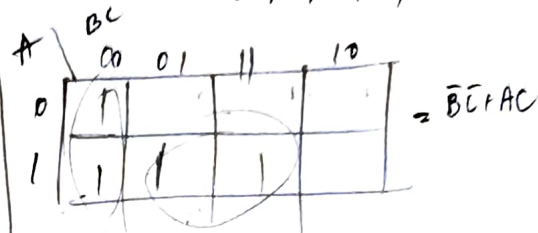
4) $F = \sum m(0,1,6,7) + d(2,3)$



5) $F = \sum m(2,3,4,5)$



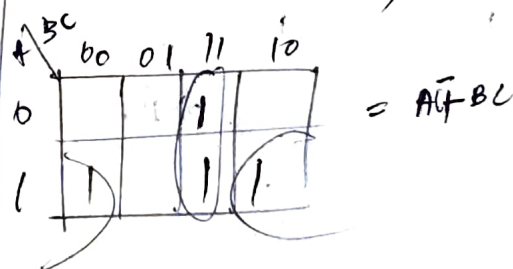
6) $F = \sum m(0,4,5,7)$



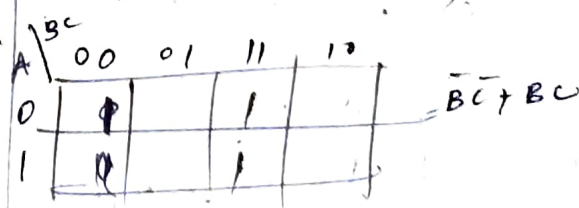
7) $F = \sum m(1,2,5,6) + d(0,7)$



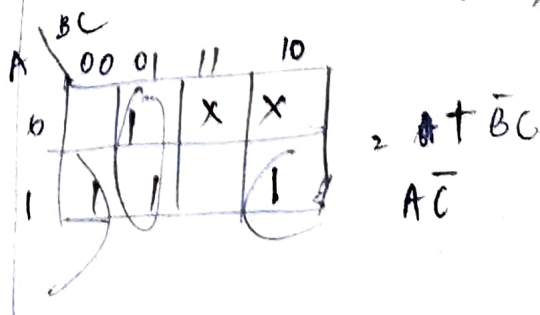
8) $F = \sum m(3,4,6,7)$



9) $F = \sum m(0,3,4,7)$

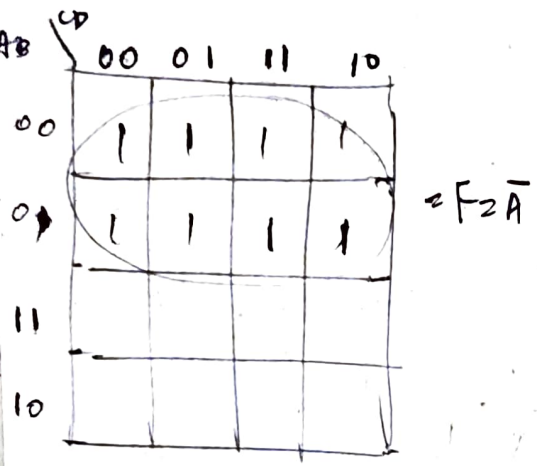


10) $F = \sum m(1,4,5,6) + d(2,3)$

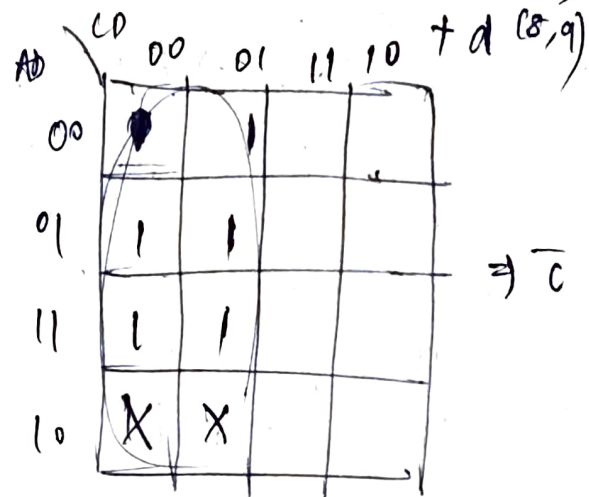


Four variable k maps

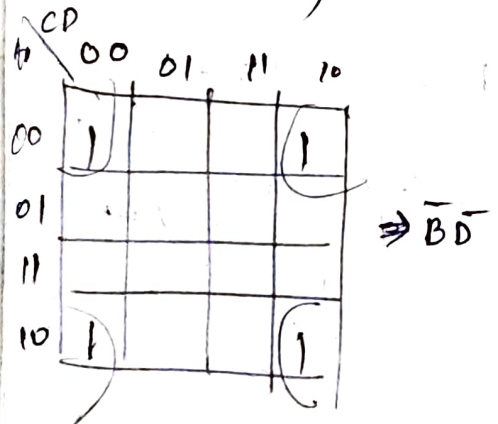
① $F = \sum m(0, 1, 2, 3, 4, 5, 6, 7)$



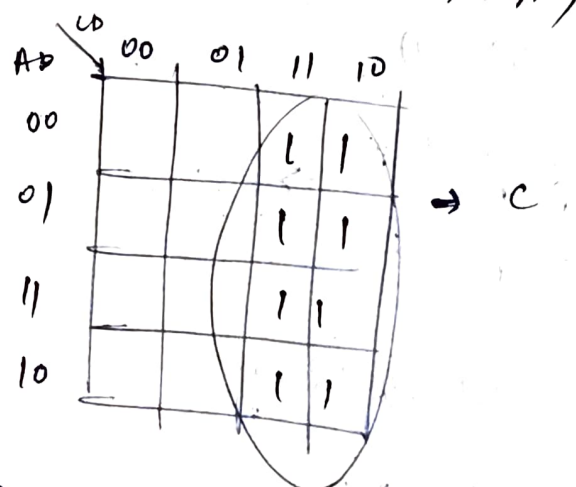
④ $F = \sum m(0, 1, 4, 5, 12, 13) + d(8, 9)$



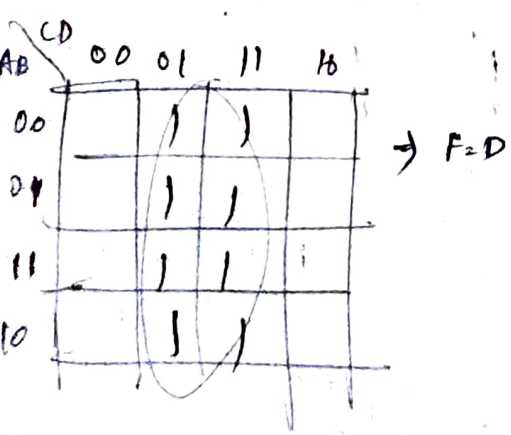
② $F = \sum m(0, 2, 8, 10)$



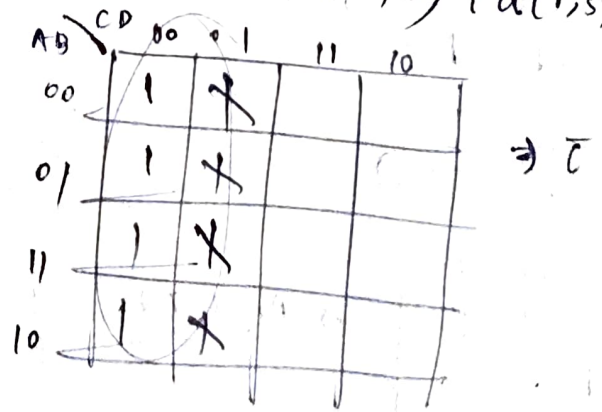
⑤ $F = \sum m(2, 3, 6, 7, 10, 11, 14, 15)$



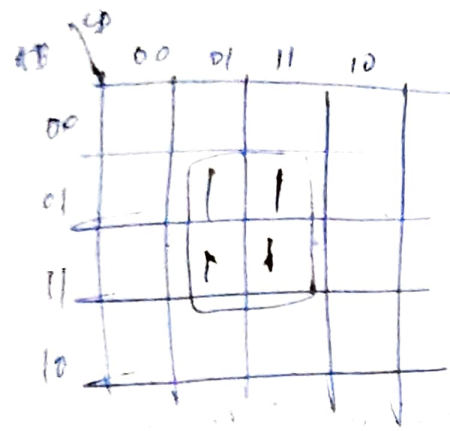
③ $F = \sum m(1, 3, 5, 7, 9, 11, 13, 15)$



⑥ $F = \sum m(0, 4, 8, 12) + d(1, 5, 9, 13)$

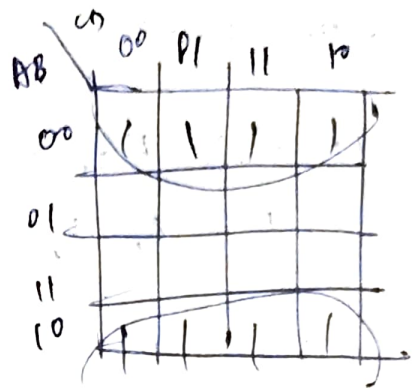


⑦ $F = \sum m(5, 7, 13, 15)$



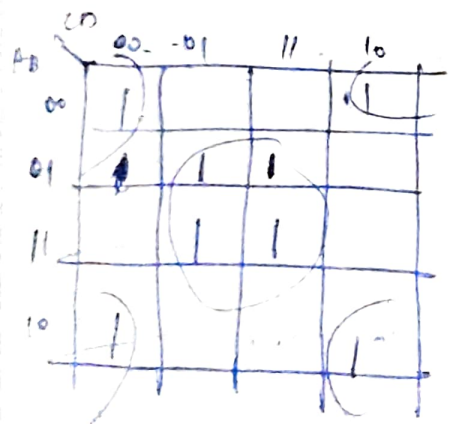
$F = BD$

⑩ $F = \sum m(0, 1, 2, 3, 8, 9, 10, 11)$



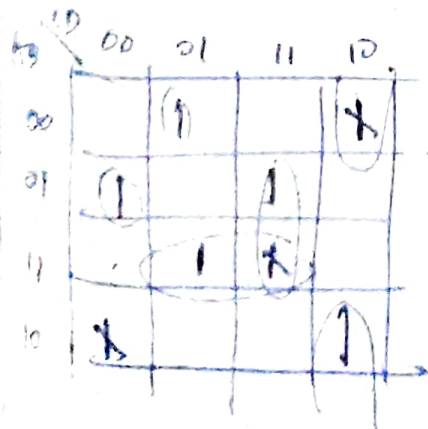
$F = \bar{B}$

⑧ $F = \sum m(0, 3, 5, 7, 8, 10, 13, 15)$



$F = BD + \bar{B}\bar{D}$

⑨ $F = \sum m(1, 4, 7, 10, 13) + d(2, 8, 15)$



$F = \bar{B}\bar{C}\bar{D} + B\bar{C}D + AB\bar{D} + \bar{A}B\bar{C}D + \bar{A}\bar{B}\bar{C}D$